

AGILE SOFTWARE ENGINEERING (CS303)

Sprint Planning & Backlog Grooming

Learning Objectives

Understand Sprint Planning, Product Backlog, Sprint Backlog, Backlog Grooming, prioritization techniques, and estimation methods used in Scrum.

Introduction

Sprint Planning and Backlog Grooming are essential Scrum activities that help teams organize work, prioritize requirements, and ensure successful sprint execution.

Sprint Planning

Sprint Planning is a Scrum event conducted at the beginning of each Sprint. The Scrum Team collaborates to define the Sprint Goal and select Product Backlog Items for implementation.

Objectives of Sprint Planning

1. Define Sprint Goal
2. Select backlog items
3. Estimate effort
4. Create Sprint Backlog
5. Plan implementation activities

Product Backlog

The Product Backlog is an ordered list of features, enhancements, bug fixes, and requirements maintained by the Product Owner.

Characteristics of Product Backlog

Prioritized, Dynamic, Transparent, Continuously Updated, and Business Value Driven.

Sprint Backlog

The Sprint Backlog contains selected Product Backlog items and the tasks required to deliver them during the Sprint.

Difference Between Product Backlog and Sprint Backlog

Product Backlog: Entire project requirements.

Sprint Backlog: Requirements selected for current Sprint.

Product Owner manages Product Backlog.

Development Team manages Sprint Backlog.

Backlog Grooming (Refinement)

Backlog Grooming is the process of reviewing, refining, estimating, and prioritizing backlog items to ensure they are ready for future sprints.

Benefits of Backlog Grooming

Improves clarity, reduces ambiguity, enhances estimation accuracy, and increases team productivity.

Prioritization Techniques

1. MoSCoW Method
2. Business Value Ranking
3. Kano Model
4. Cost-Benefit Analysis

Planning Poker

Planning Poker is an Agile estimation technique where team members use numbered cards based on the Fibonacci sequence to estimate effort.

Sprint Planning Workflow

Product Backlog → Sprint Planning → Sprint Goal → Sprint Backlog → Development → Sprint Review

Advantages

Better planning, improved collaboration, accurate estimation, reduced project risks, and higher productivity.

Exam Important Questions

1. Explain Sprint Planning.
2. Define Product Backlog and Sprint Backlog.
3. Differentiate Product Backlog and Sprint Backlog.
4. Explain Backlog Grooming.
5. Describe Planning Poker estimation.

Summary

Sprint Planning and Backlog Grooming help Agile teams prioritize work, estimate effort, and deliver high-quality software efficiently. They are key Scrum practices for successful project execution.